



SnowPro Core Certification — Complete Prep Guide

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This guide covers **100% of the SnowPro Core (COF-C02) exam** with detailed notes per domain, memory aids, 60+ practice questions with explanations, and a full mock exam.



Exam At A Glance

Item	Detail
Exam code	COF-C02
Questions	~100 (multiple choice / multiple select)
Duration	115 minutes
Passing score	~750 / 1000 (scaled)
Cost	\$175 USD
Validity	2 years
Format	Online proctored or test centre

Domain Weightings (memorise these)

#	Domain	Weight
1	Snowflake AI Data Cloud Features & Architecture	24%
2	Account Access & Security	18%
3	Performance & Cost Optimization	14%
4	Data Loading & Unloading	10%
5	Data Transformations	22%
6	Data Protection & Data Sharing	12%

 **Strategy:** Domains 1 + 5 = 46% of the exam. Master those first.



Domain 1 — Architecture & Features (24%)

The 3-Layer Architecture

1. **Cloud Services Layer** — the "brain." Authentication, access control, query parsing & optimisation, metadata management, infrastructure management. Runs on Snowflake-managed compute.
2. **Query Processing (Compute) Layer** — virtual warehouses (MPP compute clusters). Independent, elastic.
3. **Database Storage Layer** — data stored in compressed, columnar **micro-partitions** on cloud object storage (S3/Azure Blob/GCS).

Key exam fact: Storage and compute are **decoupled** — scale independently, pay separately.

Micro-Partitions

- Automatic — Snowflake creates them (50–500 MB uncompressed each).
- Columnar storage within each partition.
- Metadata stored per partition (min/max/count) → enables **pruning**.
- **Clustering** = how well data is sorted across partitions.

Virtual Warehouses

- Sizes: XS, S, M, L, XL, 2XL ... 6XL. **Each size up = 2× credits/hour** and 2× compute.
- **Multi-cluster warehouses** (Enterprise+) auto-scale OUT for concurrency (more clusters), not up.
- **Scaling UP** (resize) = bigger warehouse for complex queries.
- **Scaling OUT** (multi-cluster) = more clusters for more concurrent users.
- `AUTO_SUSPEND` / `AUTO_RESUME` control cost.

The 3 Caches (common exam trap)

Cache	Where	What it stores	Lifespan
Result Cache	Cloud Services	Full query results	24 hours
Metadata Cache	Cloud Services	Table metadata/stats	persistent
Warehouse (local) Cache	Compute	Micro-partition data	until WH suspends

Result cache serves identical queries with **zero compute** if underlying data hasn't changed.

Editions

Standard → Enterprise (multi-cluster, materialized views, 90-day Time Travel) → Business Critical (HIPAA, PCI, customer-managed keys) → VPS (isolated).

Connectivity / Ecosystem

SnowSQL (CLI), Snowsight (UI), connectors (Python, Spark, JDBC/ODBC, Kafka), Snowpark, partner tools (dbt, Tableau, Power BI).



Domain 2 — Account Access & Security (18%)

RBAC — the core model

- Privileges are granted to **roles**, roles granted to **users** (and to other roles → hierarchy).
- **System-defined roles:** `ORGADMIN`, `ACCOUNTADMIN`, `SECURITYADMIN`, `USERADMIN`, `SYSADMIN`, `PUBLIC`.
- Hierarchy: `ACCOUNTADMIN` sits on top; `SYSADMIN` + `SECURITYADMIN` under it; `PUBLIC` is inherited by everyone.
- **Best practice:** create custom roles under `SYSADMIN`; don't use `ACCOUNTADMIN` for daily work.

Access control models

- **DAC** (Discretionary): object owner grants access.
- **RBAC** (Role-Based): privileges via roles. Snowflake uses **both**.

Authentication & network

- MFA (Duo), SSO/SAML, key-pair auth, OAuth.
- **Network policies** — allow/block IP ranges.
- Federated authentication for enterprise SSO.

Data security

- Encryption **at rest** (AES-256) and **in transit** (TLS) — always on, automatic.
- **Tri-Secret Secure** (Business Critical) — customer-managed key + Snowflake key.
- **Key rotation** — automatic every 30 days.

Domain 3 — Performance & Cost Optimization (14%)

Reducing cost

- Right-size warehouses; `AUTO_SUSPEND` low (e.g. 60s).
- Use **resource monitors** with credit quotas + triggers (NOTIFY/SUSPEND).
- Prefer transient/temporary tables where Time Travel isn't needed (less storage).

Improving performance

- **Clustering keys** on very large tables (> 1 TB) with selective filters.
- **Search Optimization Service** — point-lookup queries on large tables.
- **Materialized views** — precompute expensive aggregations (Enterprise+).
- **Query Profile** in Snowsight — find bottlenecks (spilling, poor pruning).
- Warehouse **spilling to disk** = warehouse too small → scale up.

Monitoring

- `ACCOUNT_USAGE` schema (up to 365 days, ~45 min latency).
- `INFORMATION_SCHEMA` (real-time, shorter history).
- `QUERY_HISTORY` , `WAREHOUSE_METERING_HISTORY` , `STORAGE_USAGE` .

Domain 4 — Data Loading & Unloading (10%)

Stages

- **User stage** (`@~`), **Table stage** (`@%table`), **Named internal stage** (`@stage`), **External stage** (S3/Azure/GCS).

Loading

- `COPY INTO <table>` — bulk load. Uses a warehouse.
- **Snowpipe** — continuous/auto load via `AUTO_INGEST` + cloud notifications. **Serverless** (no warehouse).
- File formats: CSV, JSON, Avro, ORC, Parquet, XML.
- `ON_ERROR` options: ABORT_STATEMENT (default), CONTINUE, SKIP_FILE.
- `VALIDATION_MODE` to dry-run.

Unloading

- `COPY INTO <stage>` from a table/query → files.
- Default output = compressed (gzip) CSV; can specify format.
- `SINGLE = TRUE` for one file; default splits into multiple.

Semi-structured

- `VARIANT` column type stores JSON/Avro/etc.
- Access with `col:path.to.field::type`.
- `FLATTEN` table function explodes arrays into rows.

Domain 5 — Data Transformations (22%)

Semi-structured querying (heavily tested)

```
SELECT v:name::STRING, v:address.city::STRING,
       f.value:sku::STRING
FROM orders, LATERAL FLATTEN(input => v:items) f;
```

Estimation functions

- `APPROX_COUNT_DISTINCT` (HyperLogLog), `APPROX_PERCENTILE`, `APPROX_TOP_K`.

Useful functions

- Window functions + `QUALIFY` (filter window results).
- `PIVOT` / `UNPIVOT`.
- `LATERAL` joins.
- Date/time, string, conditional (`IFF`, `DECODE`, `CASE`) functions.

Views

- **Standard view** — logical, recomputed each query.
- **Secure view** — hides definition + underlying data (for sharing).
- **Materialized view** — stored, auto-maintained, costs storage + compute (Enterprise+).

Sampling

```
SELECT * FROM t SAMPLE (10);           -- 10% of rows
SELECT * FROM t SAMPLE (1000 ROWS);    -- ~1000 rows
```

UDFs / Stored Procedures / Snowpark

- UDFs (SQL, JavaScript, Python, Java, Scala) — return values, used in queries.
- Stored procedures — procedural logic, can do DDL/DML.
- External functions — call external APIs.

Domain 6 — Data Protection & Sharing (12%)

Time Travel

- Query/restore historical data. Retention: 1 day (Standard, default), up to **90 days** (Enterprise).
- `AT` / `BEFORE` (timestamp, offset, or statement id).
- `UNDROP TABLE | SCHEMA | DATABASE` .

Fail-safe

- **7 days** after Time Travel expires. **Snowflake-managed only** — you cannot query it; only Snowflake support can recover. Not configurable.

Cloning (Zero-Copy)

- `CREATE ... CLONE` — instant, metadata-only, no storage until data diverges.
- Clone tables, schemas, databases.

Data Sharing

- **Secure Data Sharing** — share live data, no copying. Provider creates a **share**; consumer queries it.
- **Reader accounts** — for consumers without Snowflake.
- **Snowflake Marketplace** — public datasets & data products.
- **Data Exchange** — private hub for a group of accounts.

Replication

- Database replication across regions/clouds; **failover groups** for DR (Business Critical).

Memory Aids (last-night cram)

- "Storage, Compute, Services" = the 3 layers (bottom → top).
- Fail-safe = **7 days, non-configurable, support only**.
- Time Travel = **up to 90 days (Enterprise), 1 day default**.
- Snowpipe = **serverless, continuous**. COPY = **warehouse, bulk**.
- Result cache = **24 hours, zero compute**.
- Scale UP = **bigger (complex query)**. Scale OUT = **more clusters (concurrency)**.
- ACCOUNTADMIN > SECURITYADMIN + SYSADMIN > custom roles > PUBLIC.
- VARIANT + FLATTEN = **semi-structured querying**.

Practice Test 1 — 20 Questions (answers below)

- Q1.** Which layer is responsible for query optimization? A) Storage B) Compute C) Cloud Services D) Metadata
- Q2.** How long is Fail-safe, and can it be configured? A) 1 day, yes B) 7 days, no C) 90 days, yes D) 24 hours, no
- Q3.** You need to load files continuously as they arrive in S3. Best option? A) COPY INTO on a schedule B) Snowpipe with AUTO_INGEST C) INSERT D) External table
- Q4.** Which is TRUE about the result cache? A) Lives in the warehouse B) Lasts 7 days C) Serves identical queries with no compute for 24h D) Must be enabled manually
- Q5.** To handle a spike in **concurrent users**, you should: A) Scale up (resize) B) Use a multi-cluster warehouse C) Add clustering keys D) Increase AUTO_SUSPEND
- Q6.** Maximum Time Travel retention on Enterprise edition? A) 1 day B) 7 days C) 30 days D) 90 days
- Q7.** Which role is at the TOP of the default hierarchy? A) SYSADMIN B) SECURITYADMIN C) ACCOUNTADMIN D) PUBLIC

- Q8.** Zero-copy cloning consumes storage: A) Immediately, equal to source B) Never C) Only for changed/new data D) Double the source
- Q9.** Which function gives an approximate distinct count? A) COUNT(DISTINCT) B) APPROX_COUNT_DISTINCT C) HLL_ESTIMATE D) DISTINCT_APPROX
- Q10.** A warehouse is spilling to local/remote disk. This means: A) It's too large B) It's too small for the query C) Clustering is bad D) Cache is full
- Q11.** Which view type hides both its definition and underlying data? A) Standard B) Materialized C) Secure D) Temporary
- Q12.** Snowpipe uses which compute? A) Your virtual warehouse B) Serverless Snowflake-managed compute C) Cloud Services only D) The ACCOUNTADMIN's warehouse
- Q13.** `QUALIFY` is used to: A) Filter groups like HAVING B) Filter window function results C) Qualify column names D) Grant privileges
- Q14.** Which is a semi-structured data type? A) TEXT B) VARIANT C) NUMBER D) BINARY
- Q15.** Data-at-rest encryption in Snowflake is: A) Optional add-on B) Business Critical only C) Always on, automatic D) Manual per table
- Q16.** To cap monthly credit spend, use: A) AUTO_SUSPEND B) Resource Monitor C) Query timeout D) Network policy
- Q17.** `UNDROP TABLE` works because of: A) Fail-safe B) Time Travel C) Cloning D) Replication
- Q18.** Multi-cluster warehouses are available starting in which edition? A) Standard B) Enterprise C) Business Critical D) VPS
- Q19.** Which stage is referenced by `@~`? A) Table stage B) Named stage C) User stage D) External stage
- Q20.** Secure Data Sharing copies data to the consumer: A) Yes, full copy B) No — consumer queries live data C) Only metadata D) Only on first query

✓ Answers & Explanations — Test 1

1. **C** — Cloud Services layer parses and optimizes queries.
2. **B** — Fail-safe is 7 days, non-configurable, Snowflake-support-only.
3. **B** — Snowpipe + AUTO_INGEST is the continuous-load pattern.
4. **C** — Result cache = 24h, zero compute for identical queries on unchanged data.
5. **B** — Multi-cluster scales OUT for concurrency.
6. **D** — 90 days on Enterprise.
7. **C** — ACCOUNTADMIN is top.
8. **C** — Clones only consume storage for divergent/new data.
9. **B** — APPROX_COUNT_DISTINCT (HyperLogLog).
10. **B** — Spilling = warehouse too small; scale up.
11. **C** — Secure views hide definition + data.
12. **B** — Snowpipe is serverless.
13. **B** — QUALIFY filters window-function output.
14. **B** — VARIANT.
15. **C** — Always on, automatic (AES-256).
16. **B** — Resource Monitor with credit quota.
17. **B** — UNDROP relies on Time Travel.
18. **B** — Enterprise.
19. **C** — `@~` = user stage.
20. **B** — Sharing = live query access, no copy.

Practice Test 2 — 20 Questions

- Q1.** Which cache survives a warehouse suspend? A) Local warehouse cache B) Result cache C) Neither D) Both

- Q2.** Micro-partitions are approximately what size (uncompressed)? A) 1–5 MB B) 50–500 MB C) 1–10 GB D) Fixed 128 MB
- Q3.** Which is NOT a system-defined role? A) SYSADMIN B) USERADMIN C) DATAADMIN D) ORGADMIN
- Q4.** To share data with a party that has NO Snowflake account: A) Impossible B) Create a reader account C) Email an export D) Grant them ACCOUNTADMIN
- Q5.** `COPY INTO` with `ON_ERROR = 'CONTINUE'` will: A) Abort on first error B) Skip the whole file C) Skip bad rows, load the rest D) Retry the file
- Q6.** Scaling a warehouse from Medium to Large: A) Doubles credits/hour B) Halves credits C) No cost change D) Adds a cluster
- Q7.** Which service accelerates selective point-lookup queries? A) Materialized views B) Search Optimization Service C) Result cache D) Clustering only
- Q8.** VARIANT max compressed size per row? A) 1 MB B) 8 MB C) 16 MB D) 128 MB
- Q9.** Which lets you query data "as of" 2 hours ago? A) Fail-safe B) Time Travel with AT(OFFSET => -7200) C) Cloning D) Replication
- Q10.** A Standard-edition account's default Time Travel is: A) 0 days B) 1 day C) 7 days D) 90 days
- Q11.** Which object is billed as serverless compute? A) Virtual warehouse B) Snowpipe C) Result cache D) Materialized view refresh — (trick: MV refresh IS serverless too)
- Q12.** Best practice: daily development work should use: A) ACCOUNTADMIN B) A custom role under SYSADMIN C) SECURITYADMIN D) PUBLIC
- Q13.** `FLATTEN` is used to: A) Compress data B) Explode arrays/objects into rows C) Remove nulls D) Merge tables
- Q14.** Tri-Secret Secure is available in: A) Standard B) Enterprise C) Business Critical D) All editions
- Q15.** Which stores query results for reuse across users? A) Local cache B) Result cache C) Metadata cache D) Snowpipe
- Q16.** To reduce storage from Time Travel on a staging table you rarely need to recover: A) Use a transient table B) Increase retention C) Add clustering D) Use a materialized view
- Q17.** Which is TRUE about micro-partition pruning? A) Requires manual config B) Uses partition metadata to skip data C) Only works with clustering keys D) Slows queries
- Q18.** External functions let Snowflake: A) Run on external warehouses B) Call external APIs/services C) Share data D) Replicate
- Q19.** In RBAC, privileges are granted directly to: A) Users B) Roles C) Warehouses D) Databases
- Q20.** Snowflake Marketplace provides: A) Compute credits B) Third-party/public datasets & data products C) Training courses D) Warehouses

✓ Answers & Explanations — Test 2

- B** — Result cache is in Cloud Services, survives WH suspend; local cache is lost.
- B** — 50–500 MB uncompressed.
- C** — DATAADMIN is not system-defined.
- B** — Reader accounts serve non-Snowflake consumers.
- C** — CONTINUE skips bad rows.
- A** — Each size step doubles credits/hour.
- B** — Search Optimization Service.
- C** — VARIANT max 16 MB compressed.
- B** — Time Travel AT(OFFSET => -7200 seconds).
- B** — 1 day default (Standard).
- B/D** — Snowpipe and MV refresh are serverless; primary answer **B**.
- B** — Custom role under SYSADMIN.
- B** — Explodes nested structures into rows.
- C** — Business Critical.
- B** — Result cache (shared across users on same account).
- A** — Transient tables have no Fail-safe and minimal Time Travel → less storage.

17. **B** — Pruning uses metadata to skip partitions automatically.
 18. **B** — External functions call external services/APIs.
 19. **B** — Roles (then roles → users).
 20. **B** — Marketplace = datasets/data products.
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Full Mock Exam

A **60-question timed mock** (matching the real 100-question ratio across domains) is in [mock-exam-1.md](#). Give yourself 70 minutes, no notes.



4-Week Study Plan

- **Week 1:** Domains 1 + 2 (architecture, security). Do Practice Test 1.
- **Week 2:** Domains 5 + 4 (transformations, loading). Hands-on labs.
- **Week 3:** Domains 3 + 6 (performance/cost, protection/sharing). Do Practice Test 2.
- **Week 4:** Full mock exam, review weak areas, re-take. Book the exam.



Exam-Day Tips

- Flag-and-return: don't burn time on hard questions.
 - Watch for "select TWO" / "select all" — partial credit isn't given.
 - Eliminate obviously wrong options first.
 - Trust the memory aids above for the trap questions (Fail-safe, caches, scaling).
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